

ACOUSTIC NODE CARRIER — INSPECTION PACKET [v2.1]

2-layer CNC isolation-milled · 56 x 53 mm · unplated holes · GND pour (bottom)

LAYER / ROUTING

RED = top copper (SPI bus, mic I2S, NRST, C1 decoupling AT LoRa VDD)
BLUE = bottom copper (3V3 rails + JP1 ferrite + threaded crossing, PPPS) + POUR
GRN = 0 jumper wires! BUSY = top->via->bottom copper; DI01 = all bottom copper
0 copper crossings + 0 trace-over-pad shorts (machine-verified DRC).
3V3: LD0 feeds mic/GPS with no crossing; JP1 ferrite, then ONE 0.4mm trace
threads the ESP RX<->G1 gap (0.32mm clr, ~matches the LoRa 0.37 floor).
EDGE CUTS: cut the outline + USB-C plug notch (bottom) + coil-ant notch (top)
RF KEEPOUT (v2): clear ALL bottom GND pour inside the magenta box at the ANT
feed (ANT/ANT_PAD are the 50ohm RF output) -- do not backfill copper there.

BILL OF MATERIALS (per node; blue rows are clickable -> Amazon)

U1 ESP32-S3 SuperMini - soldered FLAT via castellations (not socketed)
U2 INMP441 I2S mic - 013.9 round, 2x3 @ 2.54, rows 7mm apart
U3 ATGM336H GPS (1x5) - hangs off TOP edge [confirm PPS pad on listing]
U4 REYAX RYLR689 LoRa (LLCC68 SPI, 915MHz, 1.27mm castellated), 2pk
U5 MCP1700-3302E/T0 LD0 3V3 T0-92 (1 GND|2 VIN|3 VOUT) 2nd:HT7333-A
C1 10uF 25V MLCC 0805 at LoRa VDD (top-mount, ties to LVDD/LGND)
Cin 1uF + 10uF 0805 MLCC assortment kit (Cin/Cout, 1uF each)
C2 470uF 10V/16V LOW-ESR electrolytic (Nichicon PM) = TX-spike buffer
L1 ferrite bead 0805 ~600ohm@100MHz (TDK MPZ2012S601) on JP1 3V3 crossing
V1 1 via (BUSY). JP1=0805 via-in-pad: solder-bridge to rail; bridge pads=bypass
BAT 18650 holder (2-wire) + your CN3791 charger -> BATT-IN pads (3.0-4.2V)
HW 4x M3 standoff/screw kit + 2-layer FR1/FR4 clad stock
2nd-source LD0: HT7333-A 20pk (T0-92, same pinout)
L1 isolates ESP+LoRa HF noise from the mic/GPS supply (~20-40dB, 50-300MHz). It
does NOT fix TX droop (C2 does) or radiated GPS<->LoRa antenna coupling. Optional:
a solder bridge across the pads bypasses it (0-ohm) if it ever causes trouble.
NOTE: coin SUPERCAPS rejected -- 30-120ohm ESR drops 4-9V on the 120mA TX pulse;
a 470uF low-ESR e-cap (<0.1ohm) sources it flat.

NET MAP (matches firmware node_config.h)

GP4 Mic BCLK GP5 Mic WS GP6 Mic SD (I2S, L/R->GND = left)
GP13 MISO GP12 MOSI GP11 SCK GP10 NSS (SPI, ESP right column)
GP8 RFSW_V2 GP9 RFSW_V1 (RF switch; each pin to its outer pad)
GP7 BUSY (copper+via) GP3 DI01 (copper) GP44 NRST (top)
GP1 GPS TX(in) GP2 GPS PPS (GPS RX = no-connect)
3V3 -> mic/gps/lora VDD (C1 at LoRa VDD) GND pour -> all grounds

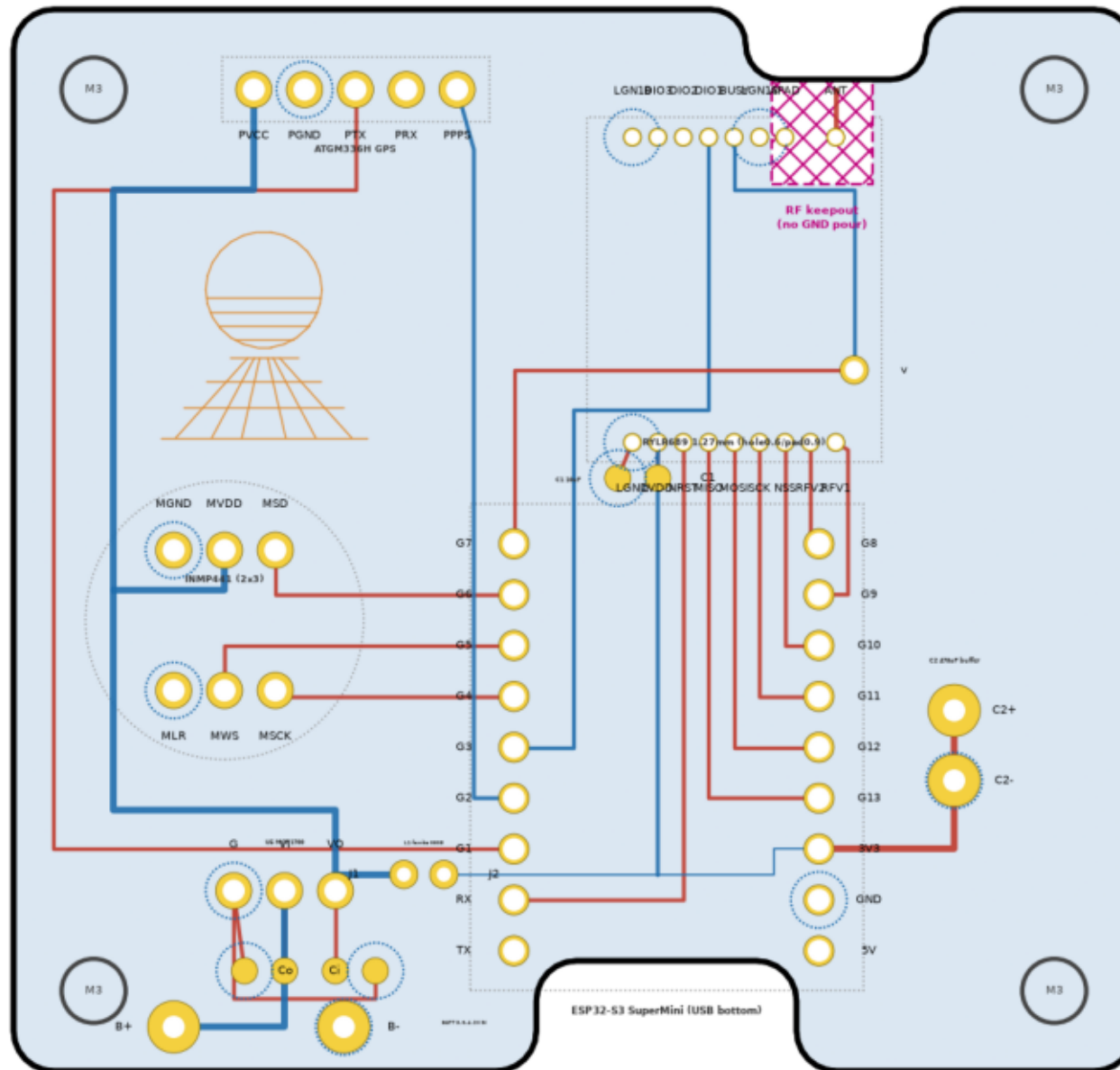
>> HOW TO CHECK PARTS FIT (before cutting copper) <<

1. Print PAGE 4 at 100% / "actual size" (NOT fit-to-page). Then caliper the 50 mm scale bar – it MUST read 50.0 mm. Re-print if off.
2. Set each real module on its pad group: pin pitch lands in the holes, the body outline (dashed) has clearance, antenna/USB edges point the right way.
3. Caliper-verify the footprints assumed vs your actual parts:
ESP SuperMini 2.54 pitch, rows 15.24, body 22.52x18
RYLR689 body 12.70x15.24, pads 1.27 pitch, rows 15.24 <- finest
ATGM336H 1x5 @ 2.54 · INMP441 013.9 round, 2x3 @ 2.54, rows 7mm
4. DRILL-ONLY test cut on scrap; dry-fit ALL parts + standoffs; THEN mill.
5. Optional: mill one board first, populate + verify before making the 2nd.

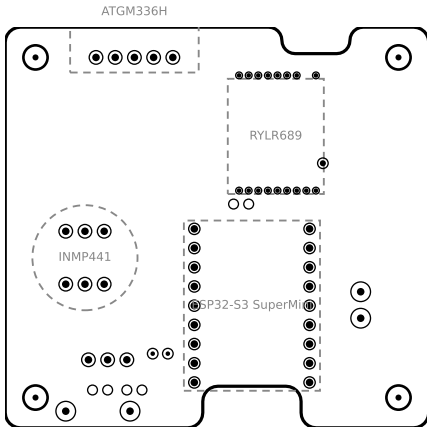
pages: 2 = placement (assembly) 3 = routed copper 4 = 1:1 fit/drill template

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RED=top copper BLUE=bottom copper GREEN dashed=jumper wire light-blue field=GND pour · 0 copper crossings



4 — 1:1 FIT / DRILL TEMPLATE (solid dots = drill points, dashed = module b



50.0 mm — PRINT AT 100%; this MUST measure 50 mm with calipers